FINAL PROJECT

Lab 1: 17/03/2026

GOAL: Setup RLC circuit for calibration

Resistors: 330 Ω, 1.1 kΩ, 3.3 kΩ

Capacitors: 5pF, 6pF, 7.5pF, 8pF, 10pF

Inductor: 46.4 mH, 10 mH (DIY)

No data was recorded since we were encountering hardware problems. The MokuGo was not reading any output

Lab 2: 18/03/2026

GOAL: Fix hardware problem with RLC and record data for calibration

The breadboard was received (see pictures) and now the MokuGo reads an output signal

However, the output did not match the theory.

Eq. used: $f_0 = \frac{1}{2\pi\sqrt{LC}}$

Using $L = 46.4$ mH, $C = 8$ pF, we found that $f_0 = 261$ kHz

Experimentally, we found that the output signal is maximized at ~ 50kHz
→ Theory mismatch

Problem: We are measuring the voltage around capacitor, not the current in the circuit. Hence no data was recorded.

Lab 3: 19/03/2026

GOAL: Finally fix RLC analysis and record first set of data for preliminary analysis

Instrument switch for MotoGo
Oscilloscope → Frequency Response Analyser
~~Waveform Generator~~

Set the input (measured voltage) at the resistor terminal and using MATH CHANNEL in the Moto measure $I = \frac{V_R}{R}$ → $R = 330\,\Omega$

Tested around with FRA and obtained a resonance curve, but not at the right peak

No data was recorded

Lab 4: 23/03/2026

GOAL: Test hypotheses and record calibration data

The reason the resonance curve was off might because of parasitic capacitance within the circuit => Build a calibration curve to account for it using following formula:

$$\frac{1}{\omega_0^2} = L \cdot C + L \underbrace{C_p}_{\text{parasitic capacitance}}$$

To be sure, test larger capacitance ($\approx 1mF$), if the resonance agrees w/ theory, then this validates the hypothesis

FRA Config!

AC coupling
Impedance 1M (In & Out)
Att. 0dB

The circuit now seems to work!
We will record data tomorrow #Tired

Lab 5: 25/03/2026

GOAL: Record calibration data

Result: Data Recorded

Lab 6: 30/03/2026

GOAL: Experimenting with new setup

Got the new thermocouple (see video for cable configuration)

Motto: Go { Power Supply 1: -5V
Power Supply 2: 10V

Don't forget to ground

Tried different configurations for capacitor in LN2 container (dewar)

↳ Broke :/
sorry

Lab 7: 01/04/2026

GOAL: Retake Calibration data

=> Noticed that using 10x probe significantly reduced parasitic capacitance

We still discarded 69.1 nF capacitor data

New task -> Get thermocouple voltage to temperature calibration plot in Python

See picture for capacitor / LN2 setup in thermos

Results: Cap calibration obtained

Lab 8: 02/04/2026

GOAL: test data collection

Attached cables to capacitor everything is connected, Problem w/ sample cap => misaligned

New task -> Code Motion: Go to be able to collect thermocouple and RLC data simultaneously

Results: Setup ready for data collection once Python is done, sample cap fixed

Lab 9: 07/04/2026

GOAL: Debug code and collect data

Code is working, make sure probe is connected to circuit and doesn't open the wire.

First test run: cool down cap until can't no more

→ managed to go down to -120°C, results to be analyzed

Lab 10: 08/04/2026

GOAL: Find what's wrong with setup

⇒ previous data is noisy and doesn't show significant structures

Test with DMM ⇒ cap plates are shorting, creating erratic capacitance

⇒ added pieces of paper between plates to avoid problem

Took noise data (for 30 points, no significant structure shown)

New data collection ⇒ able to see phase transitions, but capacitors still shorts below -30°C, limiting obtaining good data

Result: Good data obtained!

Lab 11: 09/04/2026

GOAL: Get Better data

Retook noise data.

Attempt to solder cables that are flimsy

⇒ failed since plates are aluminium

Running:

Cooling w/ sample

Cooling w/out sample

To avoid unequal displacement of capacitance
cut above part of capacitors plates
Result => capacitors broken, repeated approximation
(now there is a full piece of paper as a dielectric
between plates for sample cap)

Get data for cut sample capacitor
=> data is bad and probably unusable

Result: Limits of apparatus confirmed,
end of data collection